

ESE 5760: Semiconductor Memory Devices and Circuit Design

Final Project

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Released: 3/9/2026

Due: 4/28/2026

For the final project (totaling 40 points of the course grade), you are expected to write a report with *original* analysis and critiques on topics covered in the lectures. The aim should be that with a bit more work, these reports could be submitted to a journal, conference, or turned into a Master's Thesis.

1 General guidelines

- Given the course size, we will team into groups of 3 people. Everyone will be required to document their personal contributions to the final report, along with a spoken part of the poster day.
- Could involve design, simulation, quantitative survey/analysis, critique.
- Do not just summarize what you read in the literature. Provide your own quantitative analysis.

2 Some Project Ideas

1. We've seen several examples of $> 6T$ SRAM cells. A survey and critique of these designs; analyze tradeoffs with current technology scaling trends. A strong project would use open source simulation tools (Spice) to simulate their benefits / behaviors.
2. Simulate tradeoffs for emerging memories. There several existing simulators: https://github.com/sparsh0mittal/destiny_3d_cache. We are preparing a Pythonic port of this model which will be released next week. However, these base tools have some quirks / inaccuracies. Do a deep dive on this tool for a given memory design, find flaws, and update the tool. Run several quantitative analyses to show your improvement.
3. We've discussed many reliability metrics that are simulation based. Choose one, and do a deep dive, building up a simulation approach to evaluate different reliability design choices.
4. Memory architecture. How do you expect non-volatile memories like RRAM, STT-MRAM, FeFETs to impact a given computer architecture? Need to gather information and do the analysis in terms of speed, power, energy, area.
5. Scaling trend and challenges of DRAM from 2x nm node to 1x, 1y, 1z and beyond; discuss the technologies to realize future generations of DRAM and challenges in making these possible (this is more of perspective, but must be relatively technically deep).

6. Iso-precision performance, energy and area comparison of SRAM-based in-memory computing and digital deep neural network (DNN) accelerators.
7. Domain-specific or application-specific memory designs; new metrics for memory technology evaluations.
8. Anything from your personal research projects is strongly encouraged.
9. More ideas are encouraged based on the topics you find interesting from the lecture/reading materials.

3 Milestones

Part 1: Teaming & Proposal & Check-in (10 Points)

You are expected to briefly write about the project that you plan to work on and your methodologies (0.5 pg). Provide an outline of the various sections of the report that you plan on including in the end and a weekly Gantt chart on project completion timeline (0.5 pg).

Due date: Friday, 3/27, 2026 @ 5PM

We will do an informal midway check-in, you are expected to understand the scope and the key problems related to the specific topics/technologies that you are studying. Please submit an update of where you are in relation to the above (especially as related to the Gantt chart).

Due date: Friday, 4/10, 2026 @ 5PM

Part 2: Poster Session (10 Points)

We are going to hold a poster session to summarize your results. Each team member needs to present for 1-2 minutes on their section. We will evaluate presentations based on instructor and class feedback; namely, how well motivated was the project? How well did you present the technical details of your approach, and the key results and conclusions of your work?

You must use the Penn Poster Template (PPT or Google Slides).

Due date: Posters are *due from all teams* on **Thursday, 4/26, 2026 @ 5PM** so that they can be printed in time for **the class poster session on 4/28**. If you have a conflict with this day, you must immediately inform the instructor. Missed poster presentations will result in a grade of 0 for this portion.

Part 3: Final Writeup (20 Points)

The final report should be no more than 6 pages and must follow the IEEE Transactions Template. Reports that are substantially below this length are unlikely to receive full marks. Please provide your bios, and the in the acknowledgment section detail each member's contributions (these do not count toward the page length). Please follow the style you have seen in class readings, i.e., including an abstract, introduction, detailed methods, results, conclusion. The report may contain texts, equations, tables and figures as appropriate. Original analysis is important. List all the references using IEEE style. LaTeX & overleaf are strongly recommended.

Due date: Friday, 4/30, 2026 @ 5PM.

4 Other ground rules.

1. All policies regarding the Code of Academic Integrity apply (see the PennBook)

2. Acknowledgments: Please specify individual contributions in the report. The final project grade for the team will be the same for both team members unless there are exceptional circumstances. You can also talk to people outside the team (e.g., TA, instructors, industry experts), but please mention their contributions clearly, if significant.
3. You must consult reference literature and class notes. If you use any reference source, you must provide the full citation of the literature as if you were writing a journal article for publication.
4. Your analysis must be as quantitative as possible. Qualitative arguments are insufficient.